

Labwork 4: Neural Network

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Deep Learning – USTH

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1 Implementation

The neural network architecture is read from a text file that provides the number of layers including input, hidden, output with corresponding neurons in each layer. Three classes are defined: Neuron, Layer, and NeuralNetwork.

```
def read_network(path):
    structure = []
    f = open(path, "r")
    n_layers = int(f.readline().strip())
    for _ in range(n_layers):
        structure.append(int(f.readline().strip()))
    f.close()
    return structure

class Neuron:
    def __init__(self, weights, bias):
        self.weights = weights
        self.bias = bias

    def forward(self, inputs):
        z = sum(w * x for w, x in zip(self.weights, inputs)) + self.bias
        a = sigmoid(z)
        return z, a

class Layer:
    def __init__(self, neurons):
        self.neurons = neurons

    def forward(self, inputs):
        zs = []
        acts = []
        for n in self.neurons:
            z, a = n.forward(inputs)
            zs.append(z)
            acts.append(a)
        return zs, acts

class NeuralNetwork:
    def __init__(self, structure):
```

```

self.layers = []
for i in range(1, len(structure)):
    n_in = structure[i - 1]
    n_out = structure[i]
    neurons = [Neuron([0.0] * n_in, 0.0) for _ in range(n_out)]
    self.layers.append(Layer(neurons))

def init_random(self):
    for layer in self.layers:
        for neuron in layer.neurons:
            neuron.weights = [random.random() for _ in neuron.weights]
            neuron.bias = random.random()

def forward(self, x):
    a = x
    all_z = []
    all_a = [x]
    for layer in self.layers:
        zs, a = layer.forward(a)
        all_z.append(zs)
        all_a.append(a)
    return all_z, all_a

```

2 Results

The network is configured with structure $[2, 2, 1]$: two inputs, two hidden neurons, one output.

Hidden layer:

Neuron 1 : $w_1 = 20, w_2 = 20, b = -10$

Neuron 2 : $w_1 = 20, w_2 = 20, b = -30$

Output layer:

Neuron 1: $w_1 = 20, w_2 = -20, b = -10$

x_1	x_2	y	$\hat{y}$	pred
0	0	0	0.0000	0
0	1	1	1.0000	1
1	0	1	1.0000	1
1	1	0	0.0000	0

Table 1: Forward pass results (threshold = 0.5).

All four inputs are classified correctly.